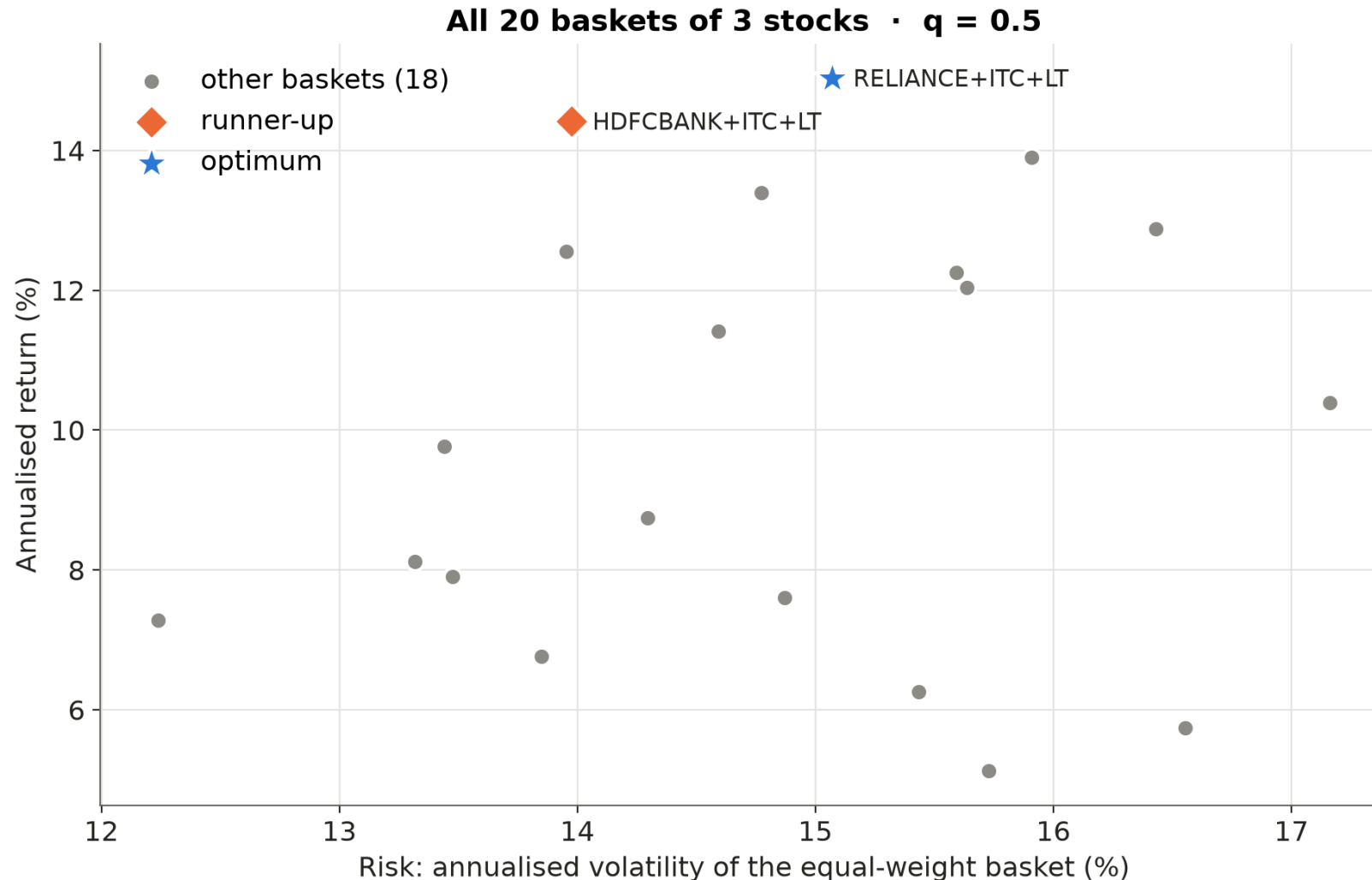


# Picking the best 3 of 6 stocks, on a quantum computer

Robo-advisor model baskets · NSE large caps · return  $\uparrow$  risk  $\downarrow$



All  $C(6,3) = 20$  baskets on the risk/return plane.

Basket count grows as  $C(n,k)$ :  
~185 000 for 10-of-20.

Goal: find the best basket for a client's risk aversion  $q$ .

## Put the constraint in the circuit, not in the cost

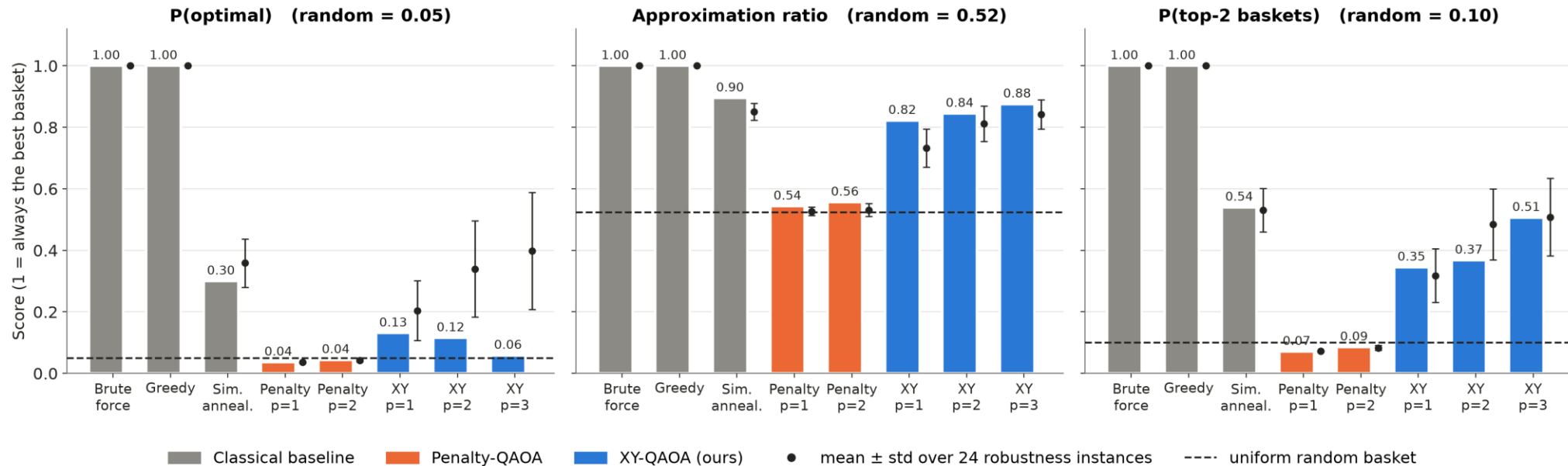
## Penalty QAOA (X mixer)

Searches all 64 bit strings, 44 invalid  
→ finds the best basket at chance level

## XY-QAOA (Dicke state + XY mixer)

Only ever visits the 20 valid baskets  
→ every shot is a real portfolio

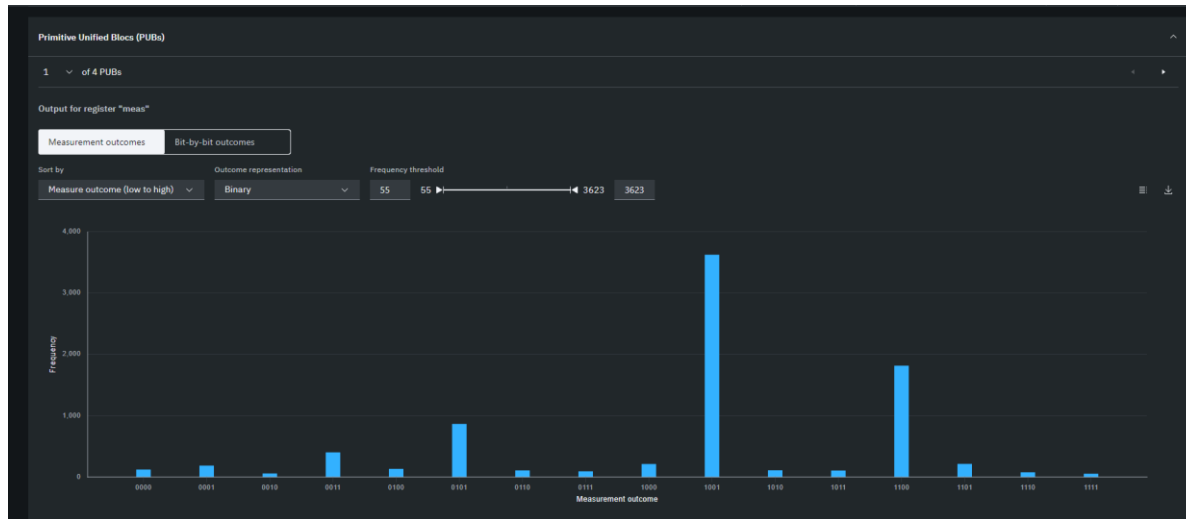
**Headline instance: pick 3 of 6 NSE stocks (RELIANCE, TCS, HDFCBANK, INFY, ITC, LT),  $q = 0.5$  · ideal simulator**



**100% valid baskets · P(optimal) 0.34 vs 0.05 random**  
**comparable to simulated annealing · penalty-QAOA = random**

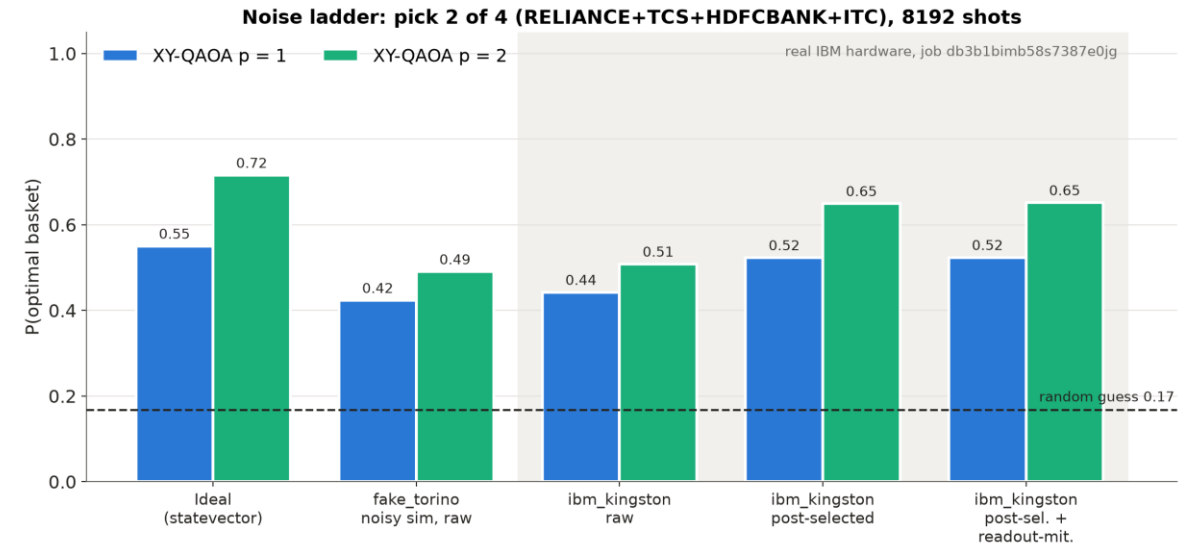
# Real IBM quantum hardware: ibm\_kingston

ibm\_kingston: 0.51 raw  $\rightarrow$  0.65 post-selected (random 0.17); top answer = true optimum



Real ibm\_kingston output — tallest bar 1001 = optimal basket (RELIANCE+ITC), second 1100 = runner-up

Limits: brute force is still faster at this size, so no quantum advantage is claimed; today's noise limits us to ~4–6 assets.



Live demo



Repo